Ah, I understand now. You are **not referring to phosphorus diffusion**, but to the **laser process used on the N-layer** during solar cell manufacturing.

There are several laser processes that act on the N-layer, depending on the cell technology.

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1. Laser Edge Isolation (Most Common)

This is the most common "N-layer laser process."

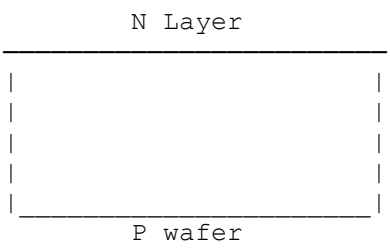
Why is it needed?

After phosphorus diffusion, the N-layer covers:

- Front surface □
- Wafer edges □

At the edge, the N-layer wraps around and contacts the P-type backside, creating an electrical short circuit.

Before isolation:

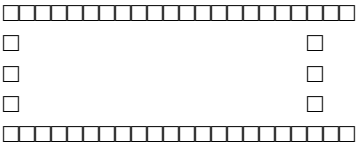


The emitter wraps around the edge.

Laser Process

A laser removes approximately **0.2–0.5 mm** of the emitter around the wafer perimeter.

After laser isolation:



```
Edge N-layer removed
```

Laser Type

Usually:

- Q-switched fiber laser
- Green laser (532 nm)
- UV laser (355 nm)

Specifications

Parameter	Typical Value
Laser Power	10–50 W
Wavelength	355 / 532 / 1064 nm
Spot Size	20–80 μm
Speed	1–3 m/s
Isolation Width	0.2–0.5 mm

Equipment Price

China:

US$80,000–250,000

Europe:

US$300,000–600,000

---

2. Laser Doping (Selective Emitter)

High-efficiency PERC/TOPCon cells often use laser doping.

Instead of removing the N-layer, the laser locally modifies it.

Process:

```
Laser

↓

Local heating

↓

More phosphorus diffuses

↓

Lower contact resistance
```

Advantages:

- Higher efficiency
- Lower series resistance
- Better fill factor

---

3. Laser Opening

Used in:

- PERC
- TOPCon
- HJT

The laser removes:

- SiNx
- Al_2O_3
- SiO_2

without damaging silicon.

Purpose:

Create contact openings.

---

4. Laser Contact Opening (LCO)

Process:

```
Passivation

↓

Laser

↓

Hole formed

↓

Metal printing
```

Very common in TOPCon.

---

5. Laser Ablation

Laser removes:

- Silicon nitride
- Oxides
- Poly-silicon

Applications:

- TOPCon
- IBC
- HJT

---

6. Laser Grooving

Older technologies used:

```
Laser
```

```
↓
Groove
↓
Metal filling
```

Less common today.

---

7. Laser Transfer Printing

Newest technology.

Laser transfers conductive paste.

Still under development.

Which Laser Process Do You Mean?

If you saw a production line labeled **"N-layer laser"** or **"N-layer laser machine"**, it most commonly refers to **Laser Edge Isolation (LEI)**.

Typical production sequence:

```
Texturing
↓
Diffusion (N Layer)
↓
PSG Removal
↓
Laser Edge Isolation
↓
PECVD
↓
```

```
Screen Printing

↓

Firing
```

Typical Chinese Equipment

Chinese manufacturers such as **DR Laser**, **Delphi Laser**, **Han's Laser**, and **HGTECH** supply laser edge isolation systems with:

Specification	Typical Value
Throughput	4,000–8,000 wafers/hour
Wafer Size	166–210 mm
Yield	>99.5%
Laser	UV or Green DPSS/Fiber
Price	**US$100,000–300,000**

One Question

Are you referring to:

1. **PERC solar cells**
2. **TOPCon solar cells**
3. **HJT solar cells**

The "N-layer laser" process is different for each technology, and I can explain the exact machine, laser type, and process flow for the specific cell architecture.